

DML COMMANDS

STEP 1 : INSERT SAMPLE DATA

The screenshot shows the SQL Developer interface. The left sidebar contains the Navigator pane with sections for MANAGEMENT, INSTANCE, and PERFORMANCE. The main window displays a query window with the following SQL code:

```
18 (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
19 (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
20 (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1800, '2024-01-13'),
21 (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
22 (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
23 (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
24 SELECT * FROM bank_transactions;
```

The Results pane shows the output of the query, displaying a table with 11 rows and 6 columns: txn_id, customer_name, branch_name, transaction_type, amount, and transaction_date.

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
104	Anil	Vizag	Deposit	8000.00	2024-01-10
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
107	Keerthi	Vijayawada	Withdrawal	1800.00	2024-01-13
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16

The bottom pane shows the Action Output, indicating that the query was executed successfully, affecting 10 rows and returning 10 rows.

QUESTION 7 – INSERT

The screenshot shows the SQL Developer interface. The left sidebar contains the Navigator pane with sections for MANAGEMENT, INSTANCE, and PERFORMANCE. The main window displays a query window with the following SQL code:

```
9 transaction_date DATE
10 );
11 INSERT INTO bank_transactions
12 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
13 VALUES
14 (101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
15 (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
16 (103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
17 (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
18 (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
19 (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
20 (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1800, '2024-01-13'),
21 (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
22 (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
23 (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
24 SELECT * FROM bank_transactions;
25 INSERT INTO bank_transactions
26 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
27 VALUES
28 (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
```

The Results pane shows the output of the query, displaying a table with 11 rows and 6 columns: txn_id, customer_name, branch_name, transaction_type, amount, and transaction_date.

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
104	Anil	Vizag	Deposit	8000.00	2024-01-10
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
107	Keerthi	Vijayawada	Withdrawal	1800.00	2024-01-13
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16
111	Venu	Vizag	Deposit	7000.00	2024-01-18

The bottom pane shows the Action Output, indicating that the query was executed successfully, affecting 10 rows and returning 10 rows.

QUESTION 8 – UPDATE

Navigator

Query 1

Limit to 1000 rows

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

12 (txn_id, customer_name, branch_name, transaction_type, amount, transactio
13 VALUES
14 (101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
15 (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
16 (103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
17 (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
18 (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
19 (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
20 (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
21 (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
22 (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
23 (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
24 • SELECT * FROM bank_transactions;
25 • INSERT INTO bank_transactions
26 (txn_id, customer_name, branch_name, transaction_type, amount, transactio
27 VALUES
28 (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
29 • UPDATE bank_transactions
30 SET amount = 5000
31 WHERE txn_id = 105;
32

```

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	21:44:50	USE bank_database	0 row(s) affected	0.000 sec
3	21:45:04	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected	0.047 sec
4	21:45:12	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transactio...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.031 sec
5	21:45:24	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
6	21:47:02	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transactio...	1 row(s) affected	0.016 sec
7	21:47:33	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.031 sec

QUESTION 9 – DELETE

Navigator

Query 1

Limit to 1000 rows

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

14 (101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
15 (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
16 (103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
17 (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
18 (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
19 (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
20 (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
21 (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
22 (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
23 (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
24 • SELECT * FROM bank_transactions;
25 • INSERT INTO bank_transactions
26 (txn_id, customer_name, branch_name, transaction_type, amount, transactio
27 VALUES
28 (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
29 • UPDATE bank_transactions
30 SET amount = 5000
31 WHERE txn_id = 105;
32 • DELETE FROM bank_transactions
33 WHERE txn_id = 111;
34

```

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	21:45:04	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected	0.047 sec
4	21:45:12	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transactio...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.031 sec
5	21:45:24	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
6	21:47:02	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transactio...	1 row(s) affected	0.016 sec
7	21:47:33	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.031 sec
8	21:48:10	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected	0.032 sec

QUESTION 10 – SELECT

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Query 1

```

29 • UPDATE bank_transactions
30 SET amount = 5000
31 WHERE txn_id = 105;
32 • DELETE FROM bank_transactions
33 WHERE txn_id = 111;
34 • SELECT *
35 FROM bank_transactions;

```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
104	Anil	Vizag	Deposit	8000.00	2024-01-10
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16

bank_transactions 2

Output

No object selected

Action Output

#	Time	Action	Message	Duration / Fetch
4	21:45:12	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.031 sec
5	21:45:24	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
6	21:47:02	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	1 row(s) affected	0.016 sec
7	21:47:33	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.031 sec
8	21:48:10	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected	0.032 sec
9	21:48:42	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

QUESTION 11 - SELECT WITH WHERE

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Query 1

```

32 • DELETE FROM bank_transactions
33 WHERE txn_id = 111;
34 • SELECT *
35 FROM bank_transactions;
36 • SELECT *
37 FROM bank_transactions
38 WHERE transaction_type = 'Deposit';

```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
104	Anil	Vizag	Deposit	8000.00	2024-01-10
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16

bank_transactions 3

Output

No object selected

Action Output

#	Time	Action	Message	Duration / Fetch
5	21:45:24	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
6	21:47:02	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	1 row(s) affected	0.016 sec
7	21:47:33	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.031 sec
8	21:48:10	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected	0.032 sec
9	21:48:42	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
10	21:49:19	SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec

QUESTION 12 - SELECT WITH ORDER BY

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Query 1

```

35 FROM bank_transactions;
36 • SELECT *
37 FROM bank_transactions
38 WHERE transaction_type = 'Deposit';
39 • SELECT *
40 FROM bank_transactions
41 ORDER BY amount DESC;

```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
104	Anil	Vizag	Deposit	8000.00	2024-01-10
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13

bank_transactions 4

Output

No object selected

Action Output

#	Time	Action	Message	Duration / Fetch
6	21:47:02	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)	1 row(s) affected	0.016 sec
7	21:47:33	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.031 sec
8	21:48:10	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected	0.032 sec
9	21:48:42	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
10	21:49:19	SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
11	21:51:12	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec